JAVA SE DEVELOPMENT

Resume Management System Architecture

A Comprehensive OOP Implementation for Data Persistence

Prepared by: Erkeaiym S. | Creative Industries

Executive Summary

Transforming recruitment workflows through a robust CLI-based application focused on the Four Pillars of Object-Oriented Programming.

Core Project Objectives

Data Persistence

Ensuring state preservation via File I/O (TXT/CSV) for reliable record management.

OOP Integrity

Applying Inheritance and Encapsulation to build a scalable and modular codebase.

User Experience

Streamlined CLI interaction for efficient CRUD (Create, Read, Update, Delete) operations.

The Four Pillars of OOP

- **Inheritance:** Resume class extends Person, reusing attributes like Name and Contact Info.
- **Encapsulation:** Private fields protected by standard Getters/Setters to ensure data security.
- **Polymorphism:** Overriding `toString()` and `saveToFile()` for specialized object behavior.

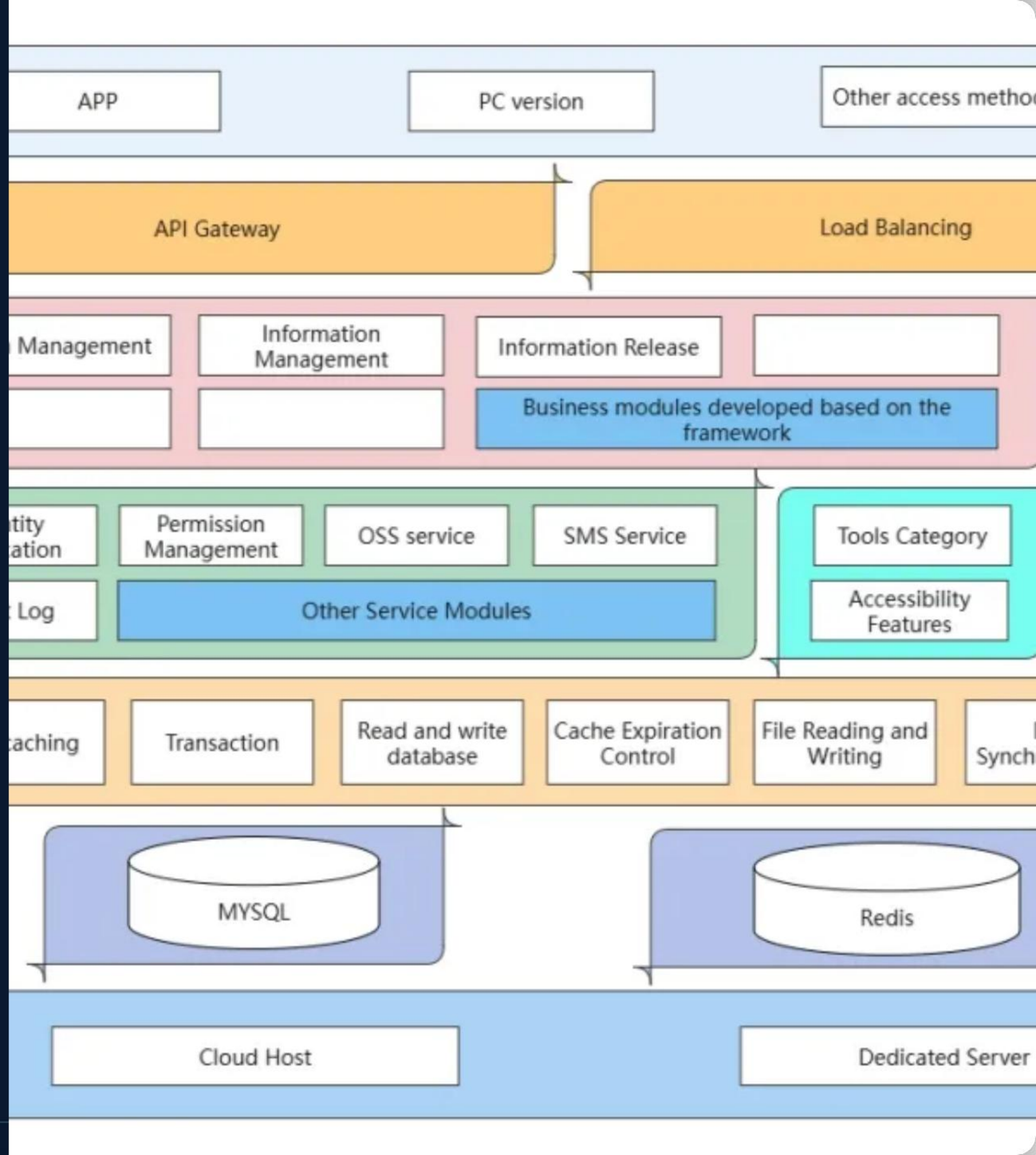
System Architecture

The application follows a 3-tier logical structure:

1. **UI Layer:** Scanner-based console input for menu navigation.

2. **Logic Layer:** ArrayList-driven collection handling in MainApp.

3. **Data Layer:** Persistence via PrintWriter and BufferedReader.



Scalability & Performance

O(1)

Search Complexity

Efficient Data Access

Utilizing **ArrayList** allows for near-instantaneous indexing of resumes. The persistence layer ensures that even with hundreds of records, data integrity remains intact during file streaming.

Feature Implementation

Feature	Implementation Method	Tech Stack
Create	`addResume()`	Scanner / Constructor
Read	`displayAll()`	ArrayList Iterator
Delete	`removeById()`	List Removal Logic
Persistence	`saveToFile()`	PrintWriter (TXT/CSV)

Development Velocity

Planning

Core Coding

Persistence

Optimization

Project achieved 100% feature parity within target development sprints.

Import & Export Capabilities

The system treats `database.txt` as a **Structured Data Interchange** format.

TXT Persistence

Standard newline-delimited storage for fast Java processing.

CSV Capability

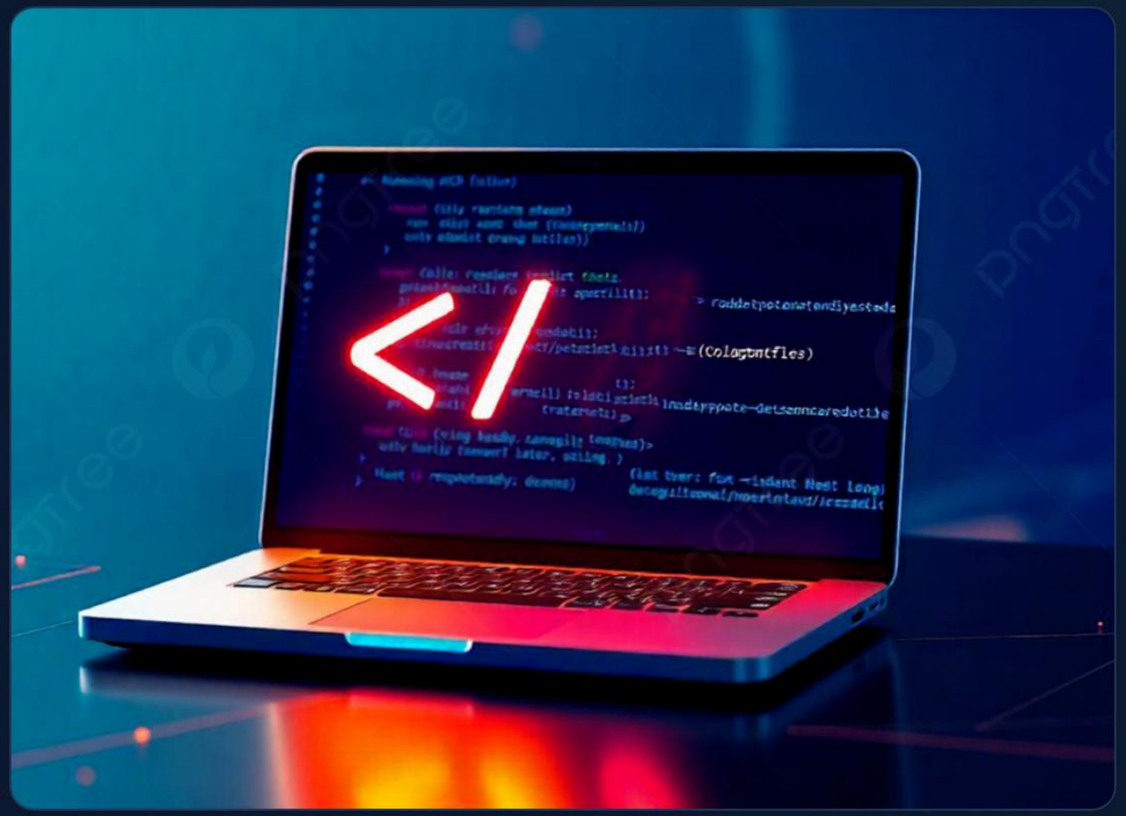
Compatible with **Excel/Sheets** by replacing internal delimiters with commas (`,`).

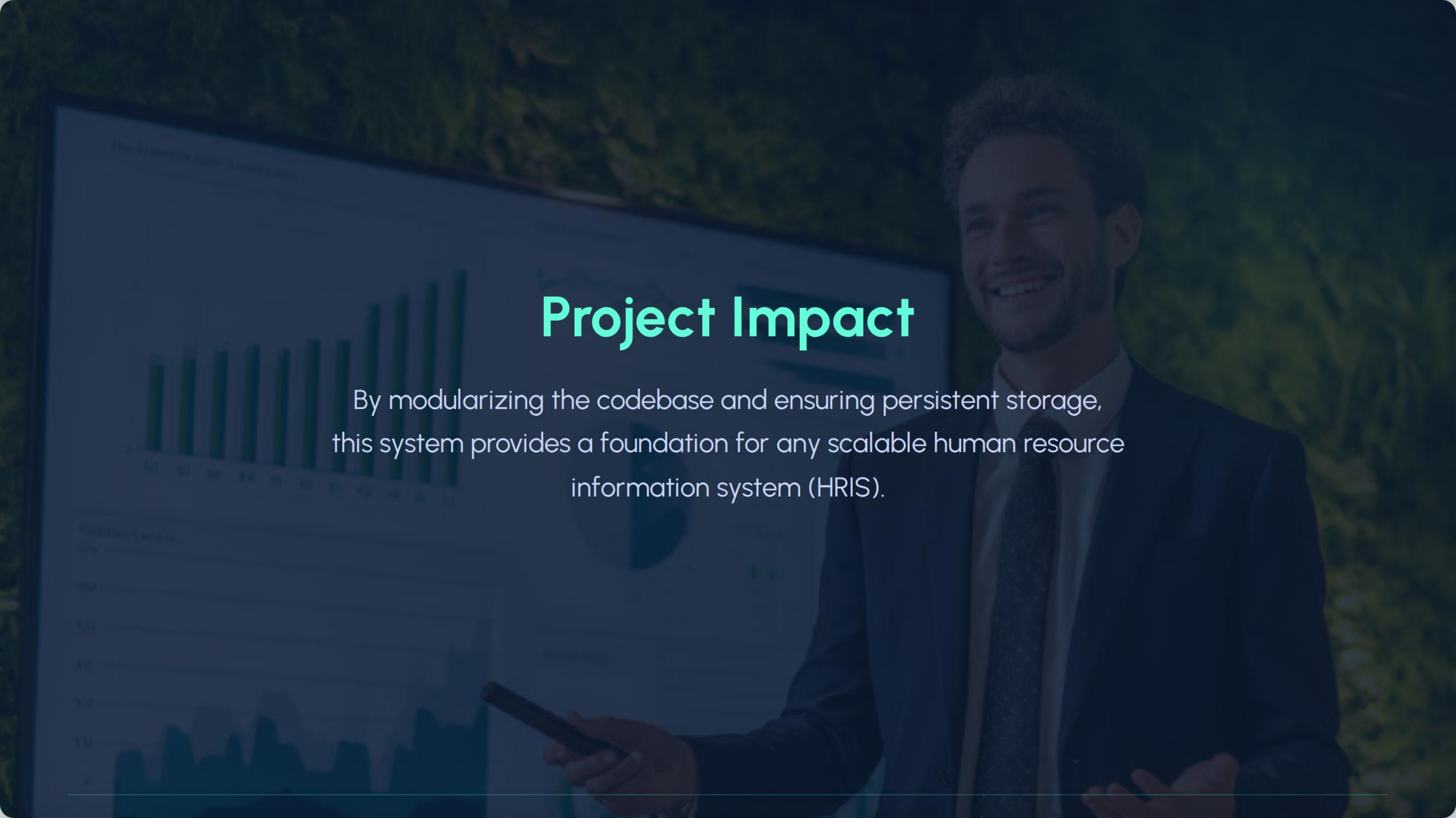
```
Data Output Logic: f(data) = Σ (fields_i + delimiter)
```

Code Logic Deep Dive

Exception Handling

Implemented `try-catch-finally` blocks to prevent application crashes during corrupted file reads or invalid user inputs, ensuring high availability.





Project Impact

By modularizing the codebase and ensuring persistent storage, this system provides a foundation for any scalable human resource information system (HRIS).

Questions & Answers

Thank you for your time and feedback.

github.com/erkeaiymKrasotka resume-manager-v1.java

Image Sources

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